Homework 1 – ECS 220, Winter 2025

Instructions

Unfortunately, you will have to submit PDFs with your solutions in both Gradescope and Canvas. This is because we will grade them in Gradescope, but we will use Canvas for peer review. Your submission to Gradescope is the official one that will be graded and that will determine the late penalty if it is late. The Canvas submission will only be used for sharing PDFs for peer review.

Please turn in *separate* PDF files (produced by LaTeX) with solutions to each of the following problems. In peer review, I prefer if the matching of submissions to people for peer review is independent on each problem, rather than reviewing the same two groups for all problems. To facilitate this in Gradescope and Canvas, each of these problems is a separate assignment, with names like "HW 1(a)", "HW 1(b)", etc. (Though all assignments link to this same PDF document with the problem statements.) So please make different PDF files, each with a solution to a *single* problem, uploaded to the different Canvas assignments (i.e., please do not just produce a single PDF with all the solutions and upload it to all the Gradescope/Canvas assignments).

To keep peer review anonymous, please do *not* include any group member's name in the PDFs.

Although the assignment may be done in groups, peer review is to be done individually.

You may work in groups of 2 or 3, or you may work on your own. See the course syllabus for more details.

Exercises

Exercise a. *Polynomial subroutines. (Textbook problem 2.19)*

Here is the exercise as stated in the textbook:

> Another sense in which P is robust is that it allows one program to use another as a subroutine. If an algorithm B runs in polynomial time, and an algorithm A performs a polynomial number of steps, some of which call B as a subroutine, show that A also runs in polynomial time.

Furthermore assume that B returns a Boolean, i.e., it also solves a decision problem. (Though not required, if you like, discuss how this could be false if B, although a polynomial-time algorithm, is allowed to return a string rather than a Boolean.)

Exercise b. *Coloring, with the oracle's help. (Textbook problem 4.2)*

> suppose we have an oracle for the decision problem GRAPH k-COLORING. Show that by asking a polynomial number of questions, we can find a k-coloring if one exists.

Hint: You want to iteratively compute a coloring, where partway through, some nodes have colors assigned already and others do not. You need to ask the oracle about a modified version of the original graph to learn if this partial coloring can be finished; if not, make a different choice when coloring the next node.

Exercise c. *Small witnesses are easy to find*

The definition of NP requires that the number of bits of the witness is at most polynomial in the number of bits of the input, i.e., $|w| = \mathsf{poly}(n)$. Suppose a decision problem $A \in \mathsf{NP}$ has the property that witnesses, when they exist, are at most logarithmic size, i.e., $|w| = O(\log n)$. Show that this implies $A \in \mathsf{P}$.